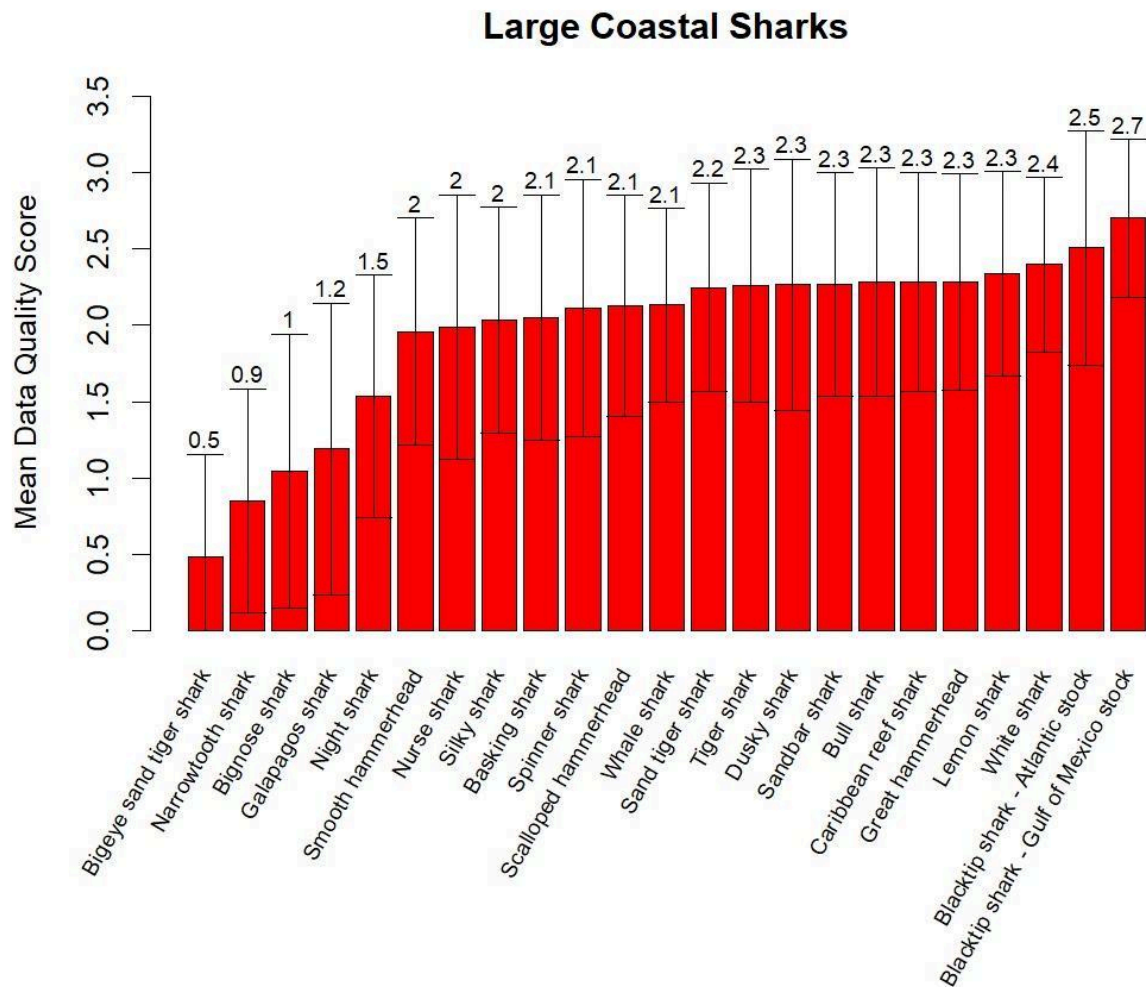


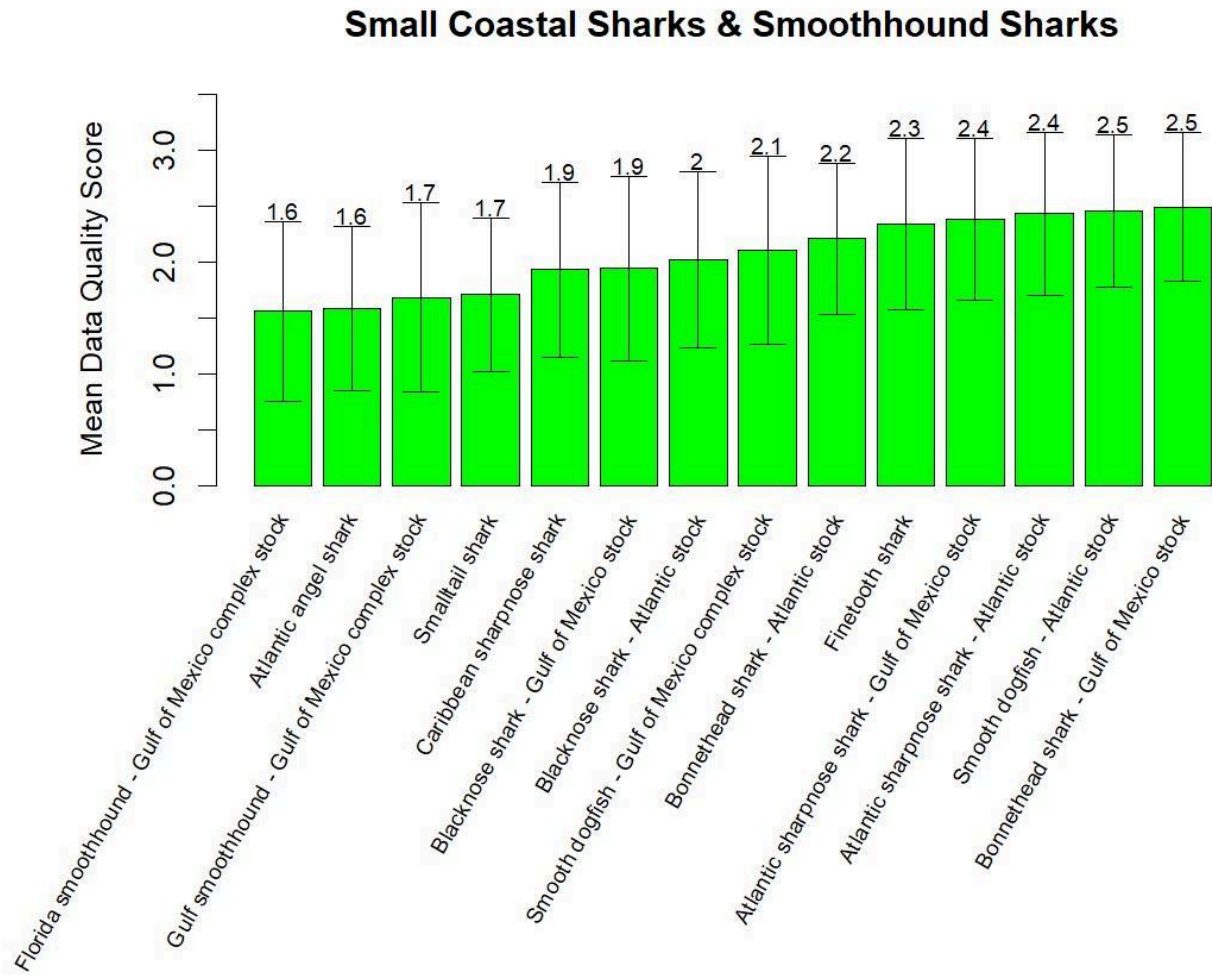
S2 File Supporting Information: Supplementary Figures

Supplementary Fig 1a-e. Mean data quality scores for each species and stock, sorted by complex. Data quality scores are on a scale of 0 to 3, with 0 being low data quality and 3 being high data quality. Error bars and numbers above each bar represent the standard deviation and mean data quality scores, respectively, for each species.

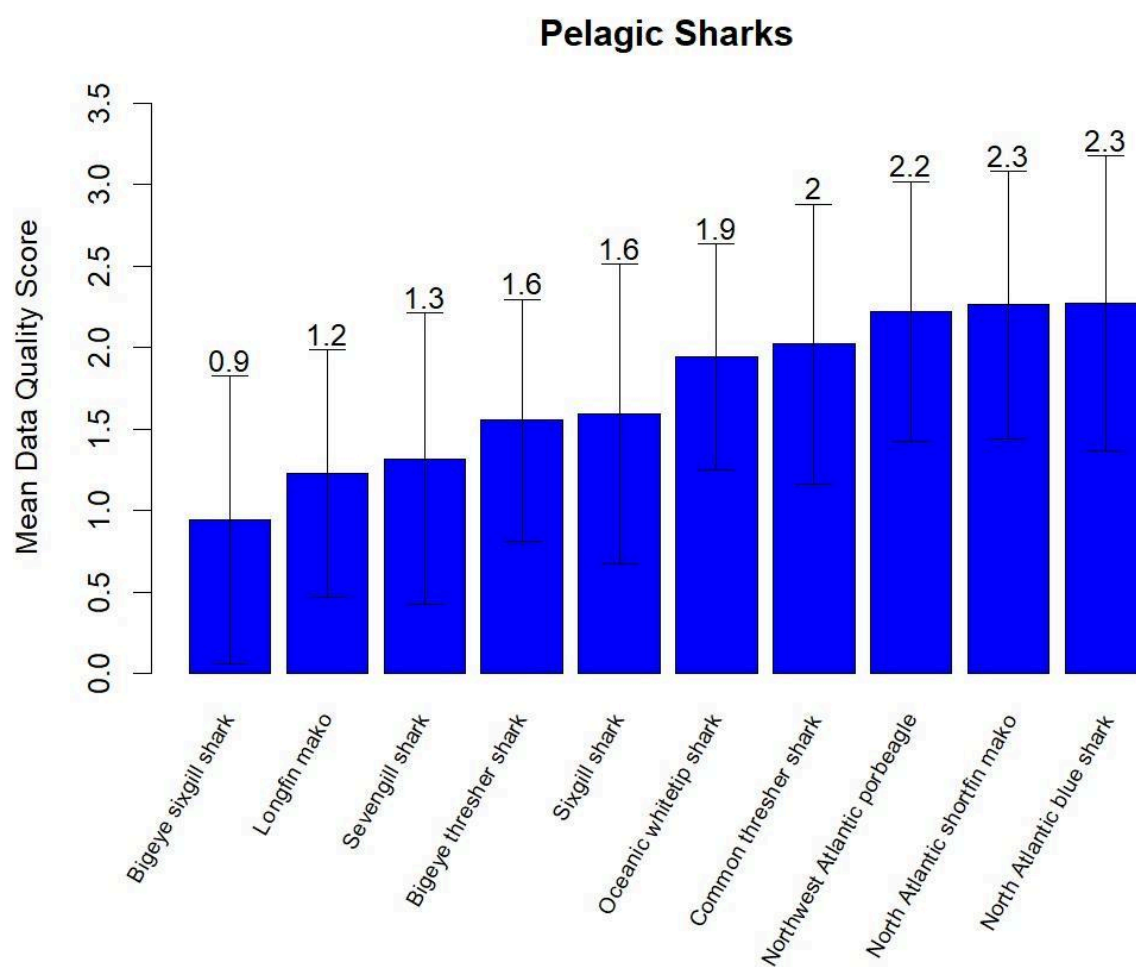
1a.



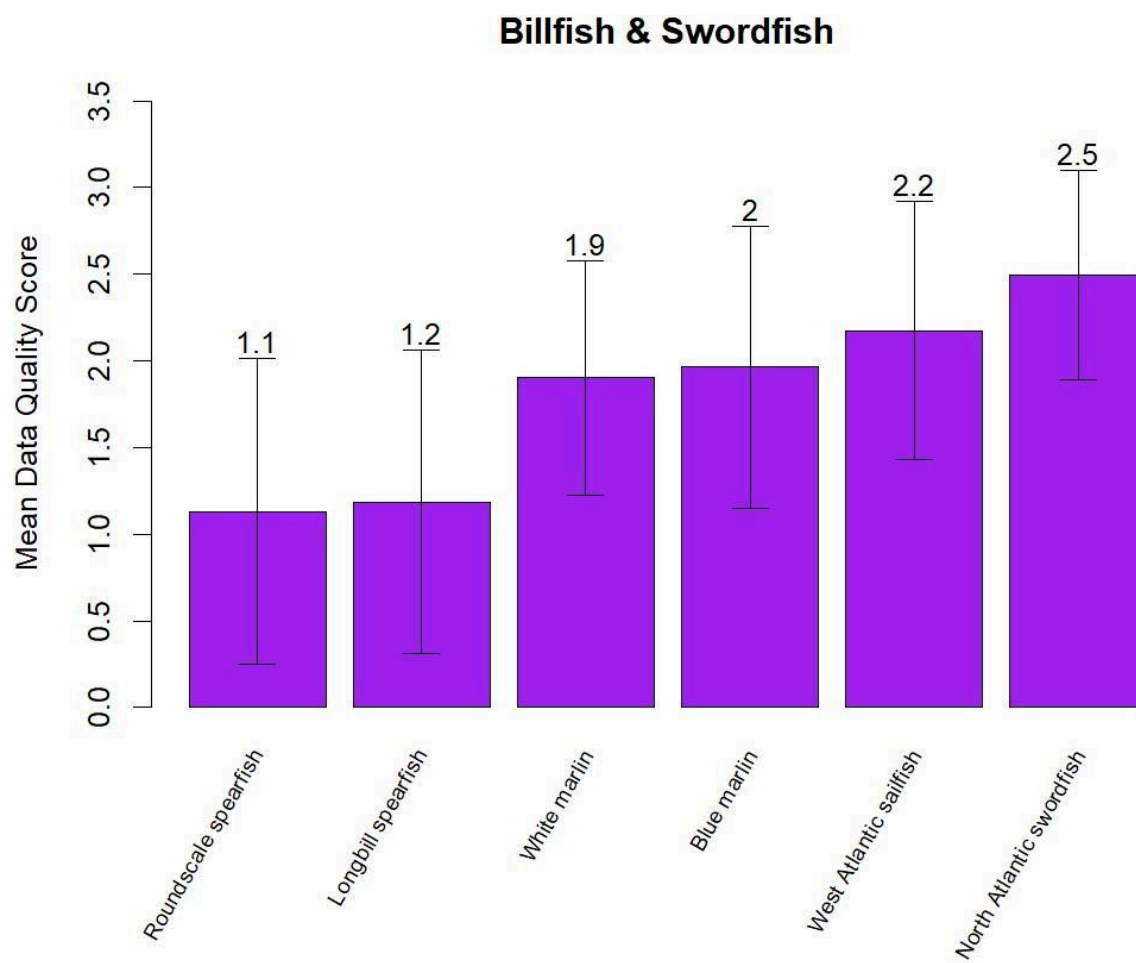
1b.



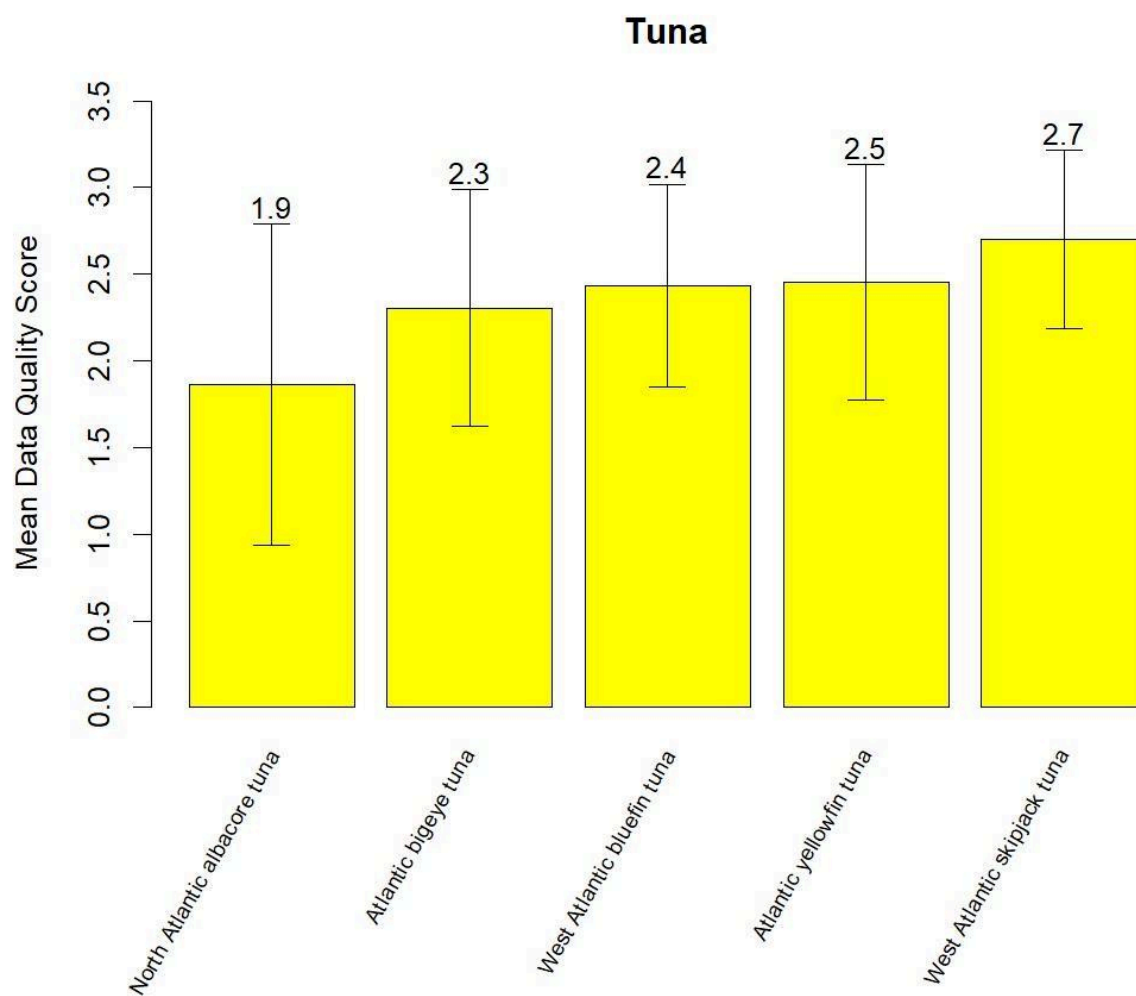
1c.



1d.

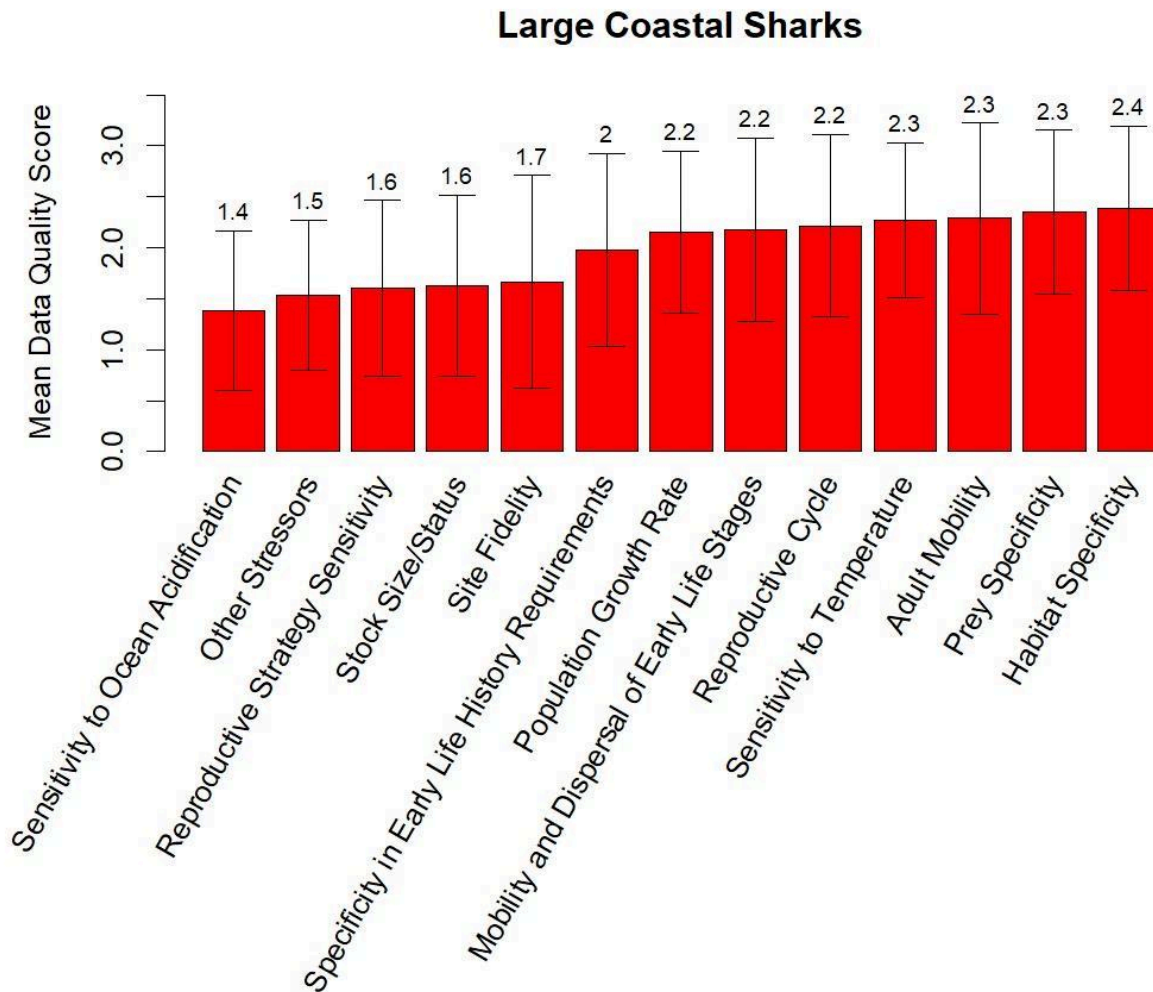


1e.

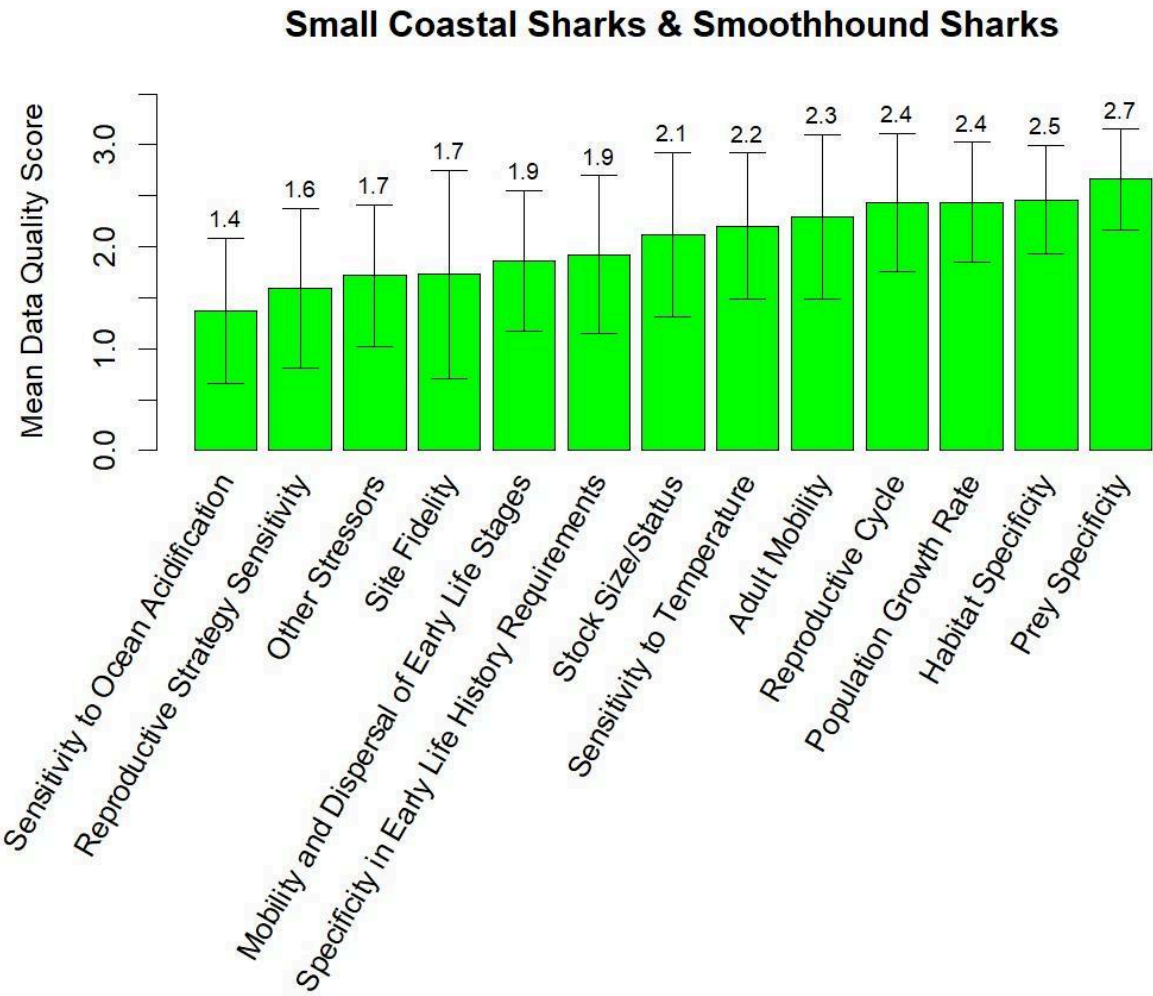


Supplementary Fig 2a-e. Mean data quality scores for each BSA, sorted by complex. Data quality scores are on a scale of 0 to 3, with 0 being low data quality and 3 being high data quality. Error bars and numbers above each bar represent the standard deviation and mean data quality scores, respectively, for each BSA.

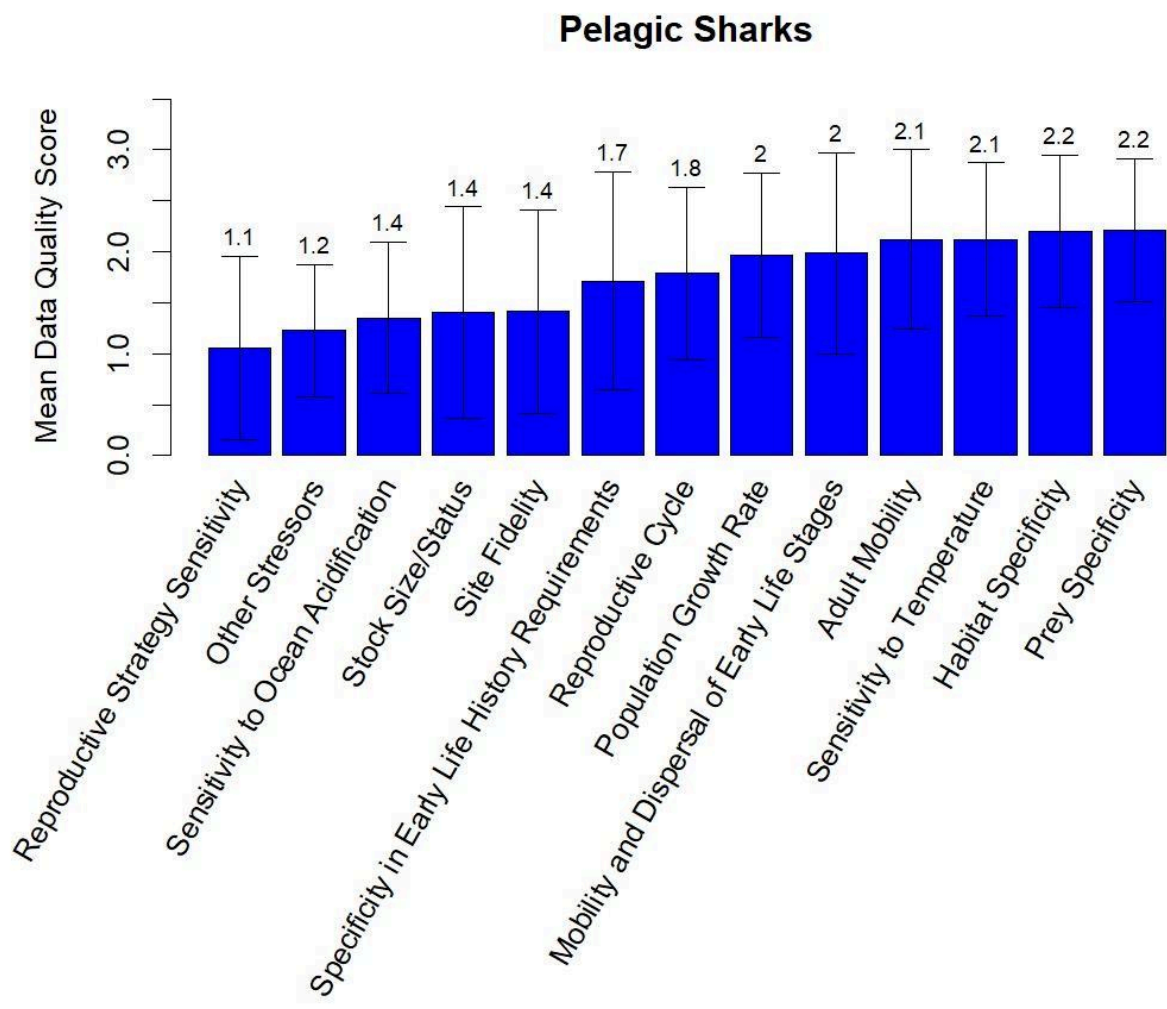
2a.



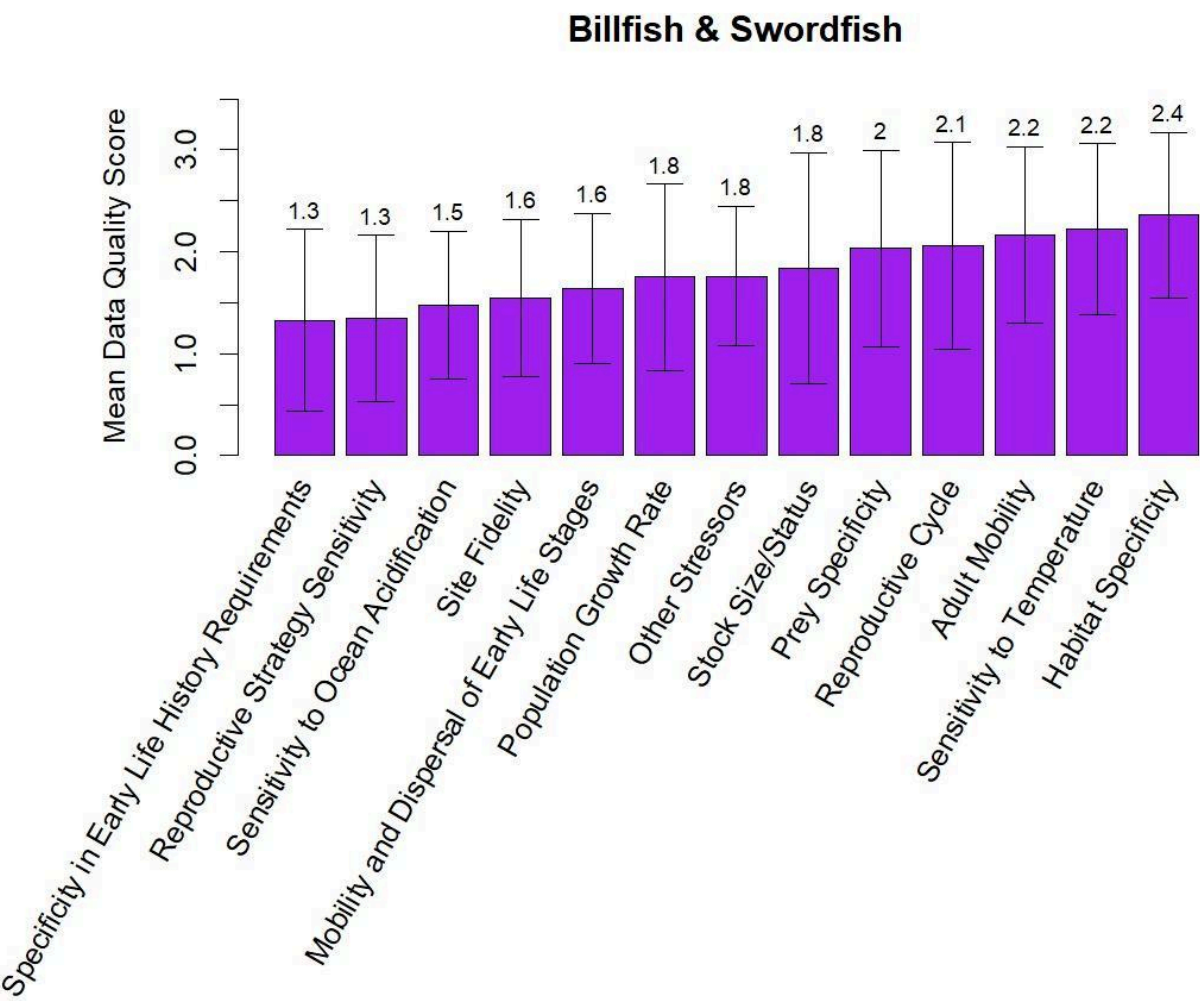
2b.



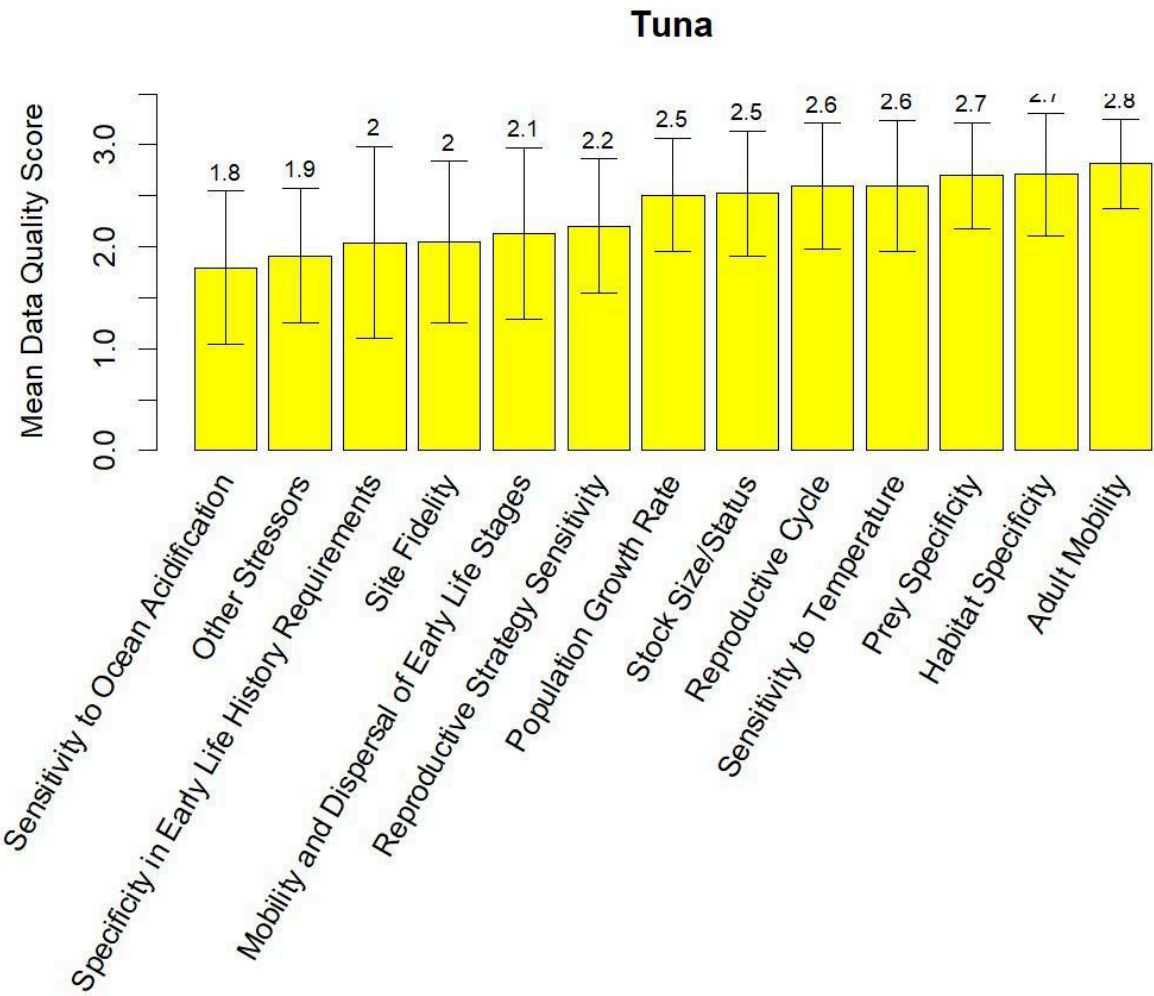
2c.



2d.



2e.



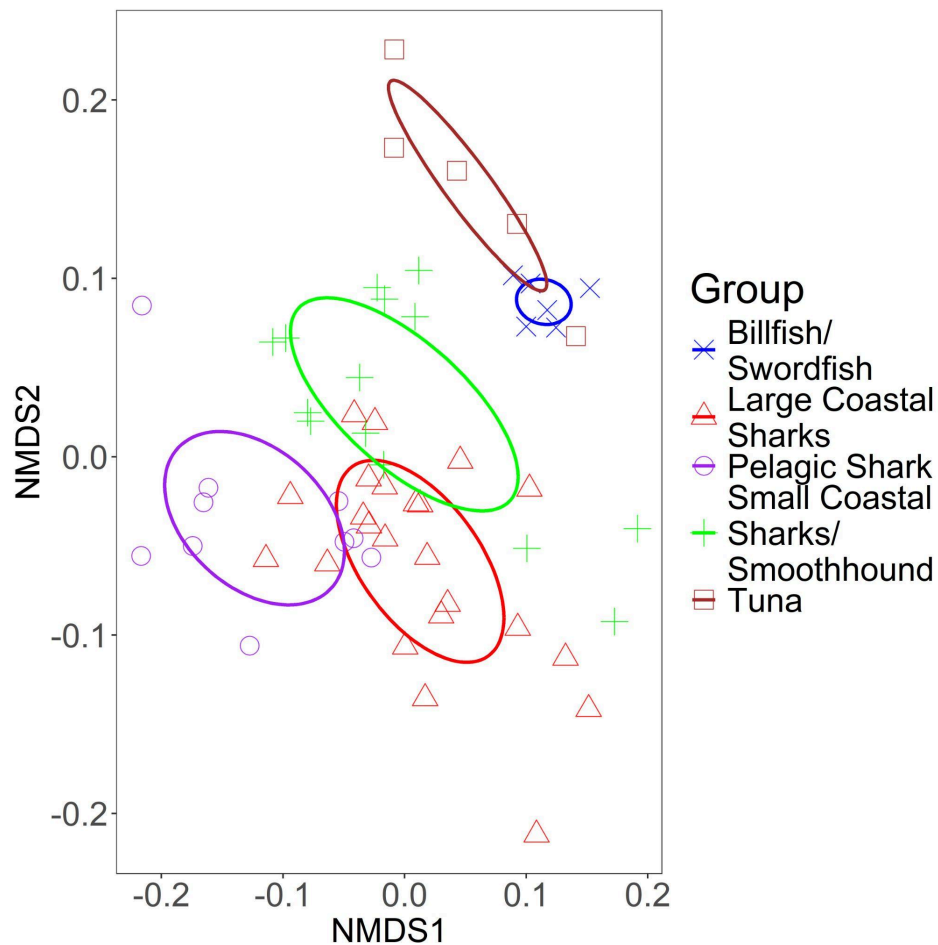
Species and stock complex evaluation

Non-metric multidimensional scaling (NMDS) was conducted using the vegan package [1] to identify whether BSA scores across species could be grouped by species and stock complex. Bray-Curtis dissimilarities were calculated among all sensitivity scores for all species. From the dissimilarity matrix, NMDS calculates and ranks relative distances and reduces the dimensionality of the data into two dimensions which allows for easier visualization. In the NMDS plot on the two dimensions, the position of each data point (representing each species and its associated BSA scores) reflects its relative similarity to other data points (BSA scores of other species), where the closer two points are to each other the more similar they are (species with similar BSA scores). Analysis of similarity (ANOSIM) was used to determine if there was a significant difference in BSA scores among species complexes. A separate ANOSIM was conducted to determine whether BSA scores across species were similar among species vulnerability rank.

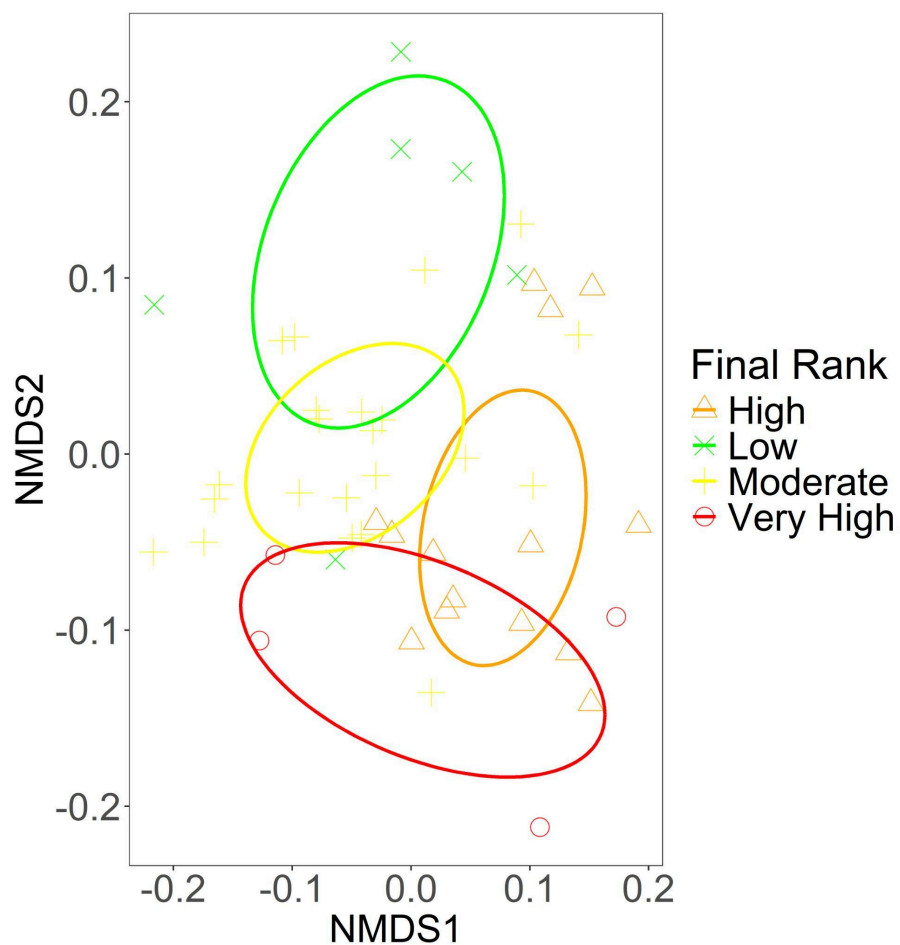
The results of the NMDS multivariate analysis are shown in Supplementary Fig 3a and 3b. Distances between points represent relative similarity in BSA scores among species and stocks. In other words, markers that are closer together represent species and stocks with greater similarity in their BSA scores than those that are farther apart. The stress value for the NMDS analysis was 0.162. ANOSIM results indicated species and stock BSA scores were more similar within the species complex ($R=0.50$; Supplementary Fig 3a) than when grouped by final vulnerability rank ($R=0.34$; Supplementary Fig 3b).

Supplementary Fig 3a-b. Non-metric multidimensional scaling (NMDS) plots of scores for 58 species and stocks. Points represent individual species and stocks and their associated BSA scores. The stress value for the NMDS analysis was 0.162. (a) Plot of points colored and symbolized by species complex. (b). Plot of points colored and symbolized by vulnerability rankings. Ellipses represent the standard deviation around each grouping level (e.g., Large Coastal Sharks). Analysis of similarity results indicated species and stock BSA scores were more similar within the species complex ($R=0.50$; Fig 3a) than when grouped by final vulnerability rank ($R=0.34$; Fig 3b).

3a.



3b.



References

1. Oksanen J, Guillaume FB, Friendly M, Kindt R, Legendre P, McGlinn D, et al. vegan: Community Ecology Package. R package version 2.5-72020. Available from: <https://CRAN.R-project.org/package=vegan>.